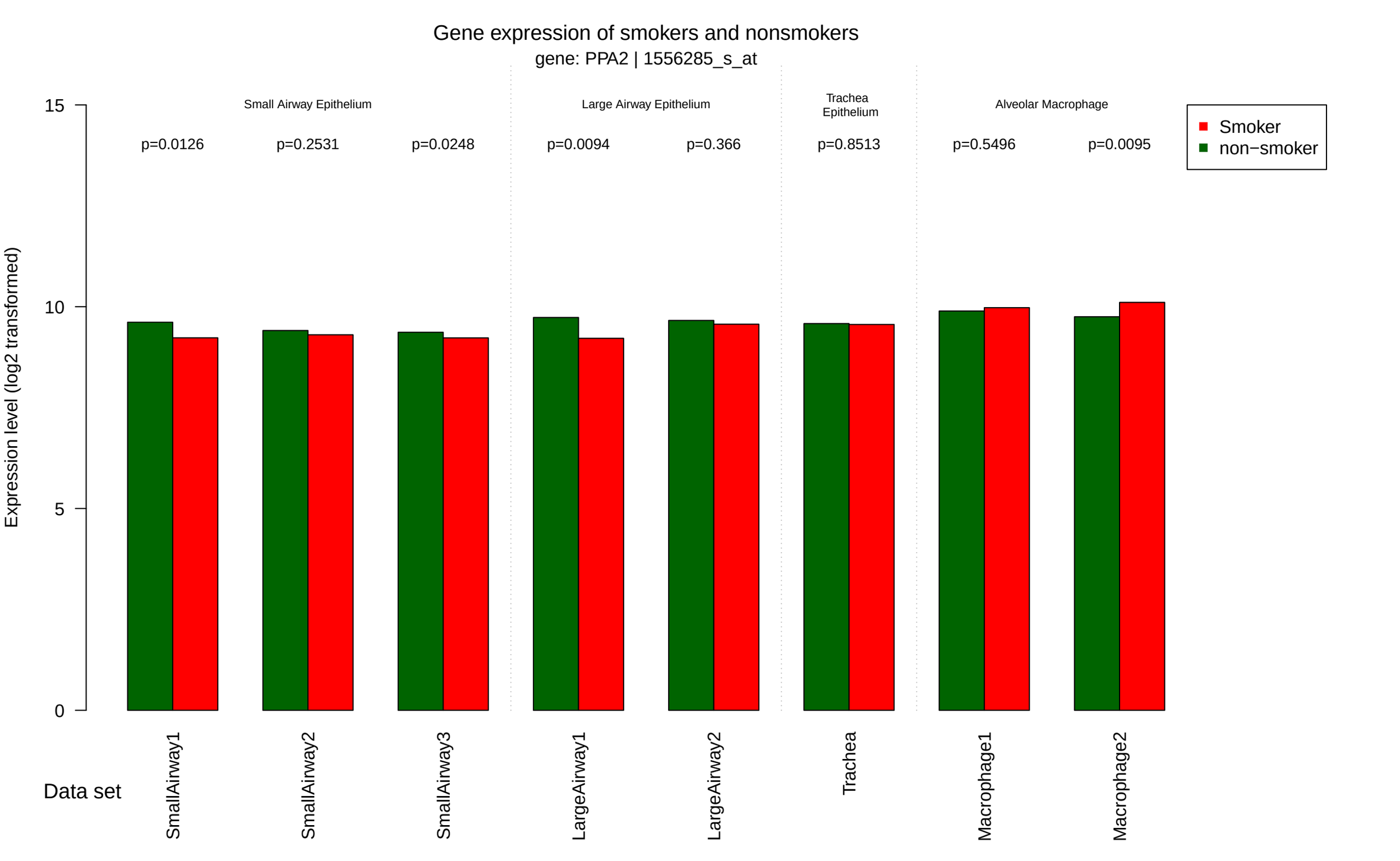
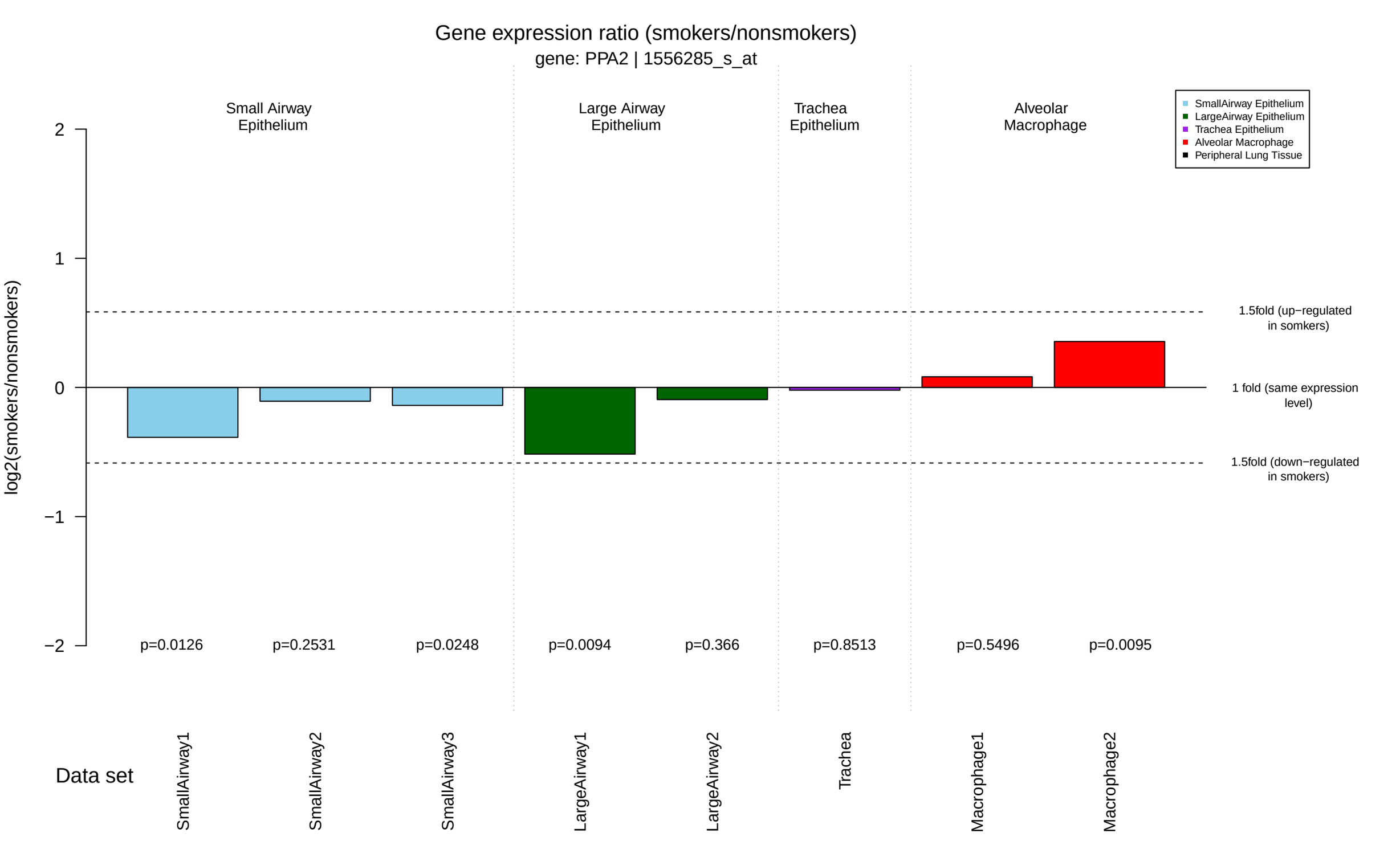
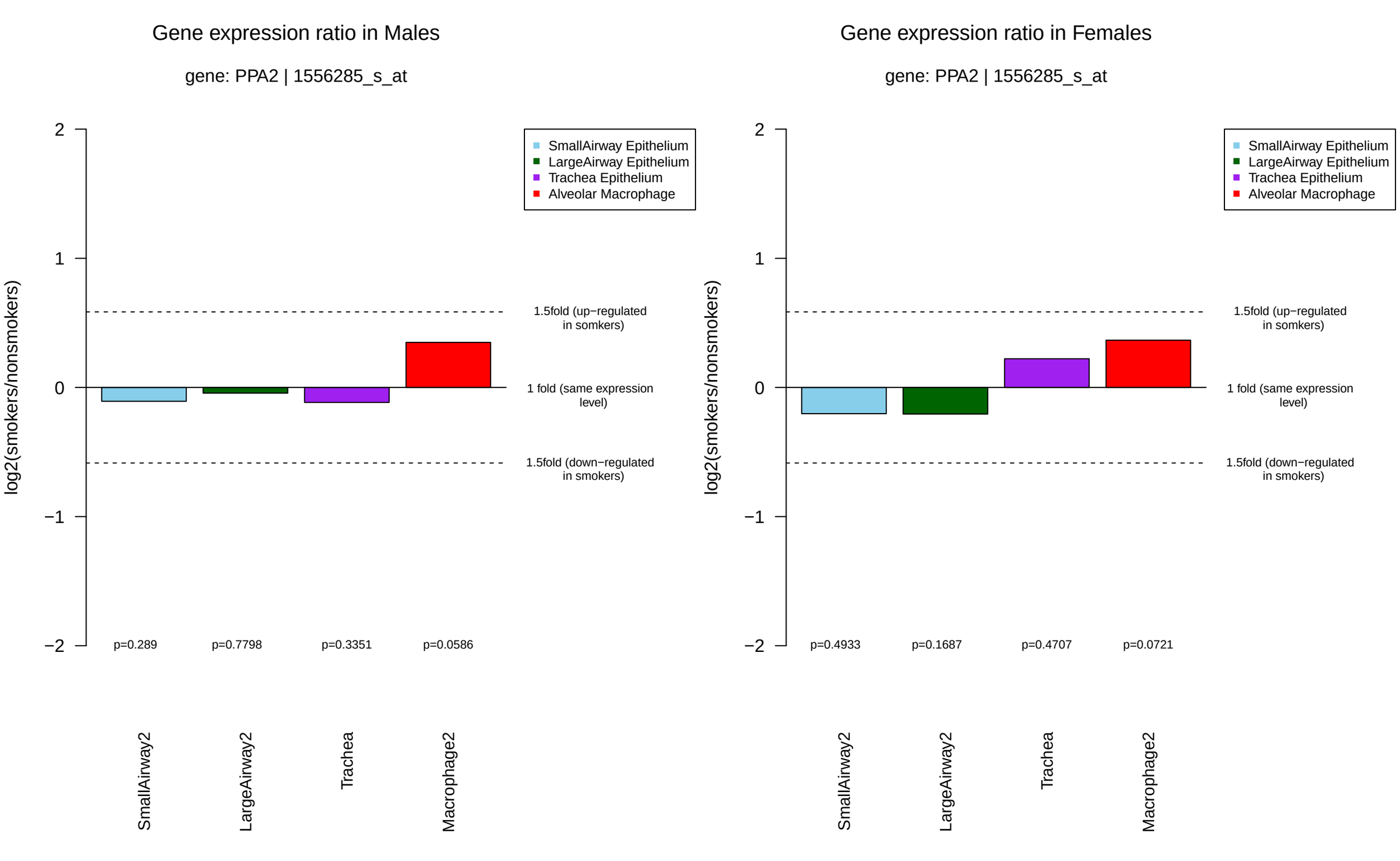
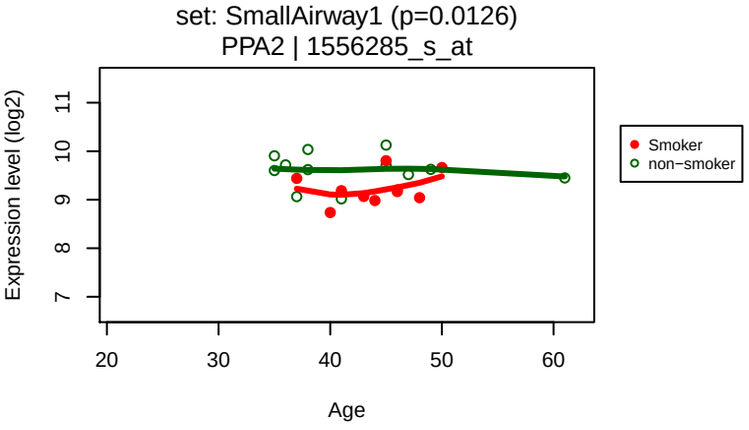
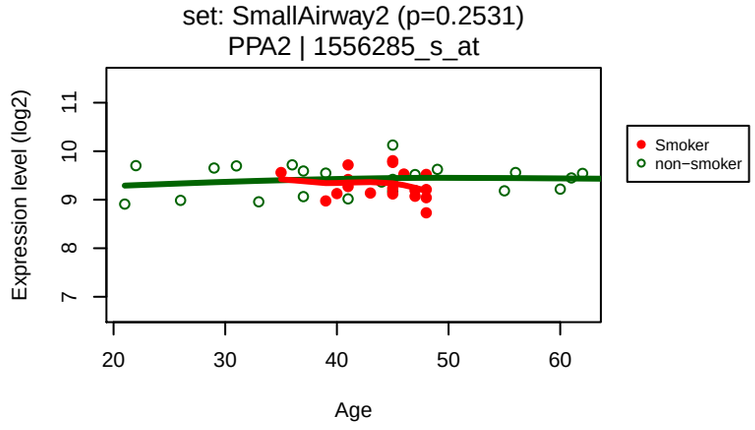
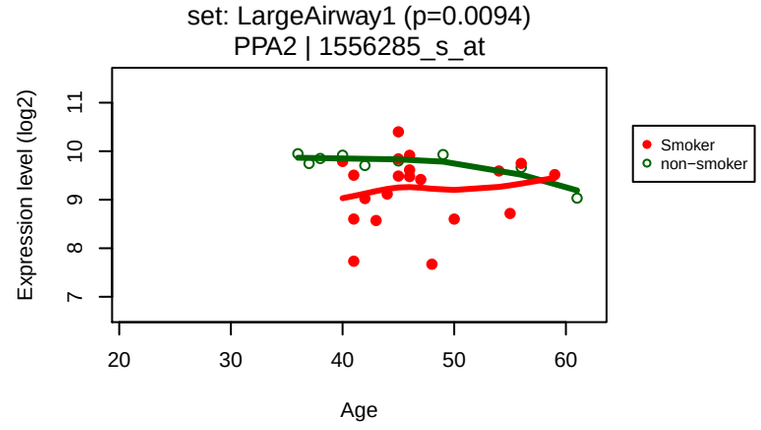

Gene expression of smokers and nonsmokers
gene: PPA2 | 1556285_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0126
p=0.2531
p=0.0248
p=0.0094
p=0.366
p=0.8513
p=0.5496
p=0.0095
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

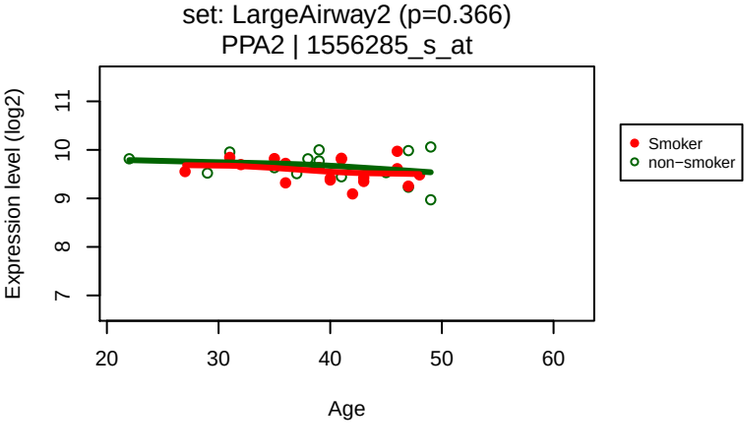

Gene expression ratio (smokers/nonsmokers)
gene: PPA2 | 1556285_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0126
p=0.2531
p=0.0248
p=0.0094
p=0.366
p=0.8513
p=0.5496
p=0.0095
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

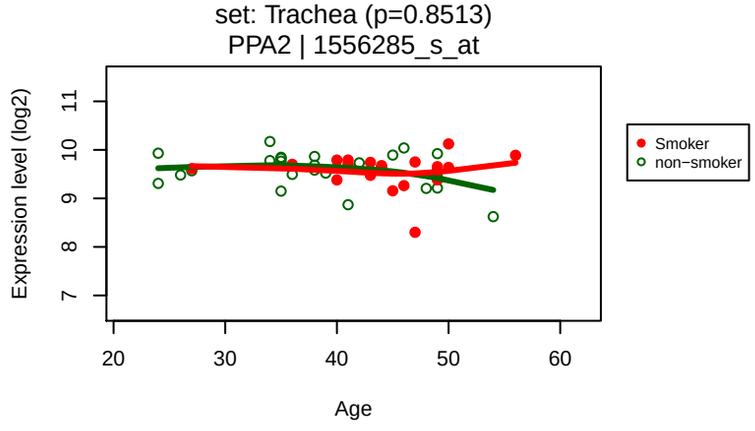

Gene expression ratio in Males
gene: PPA2 | 1556285_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.289
p=0.7798
p=0.3351
p=0.0586
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPA2 | 1556285_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4933
p=0.1687
p=0.4707
p=0.0721
SmallAirway2
LargeAirway2
Trachea
Macrophage2

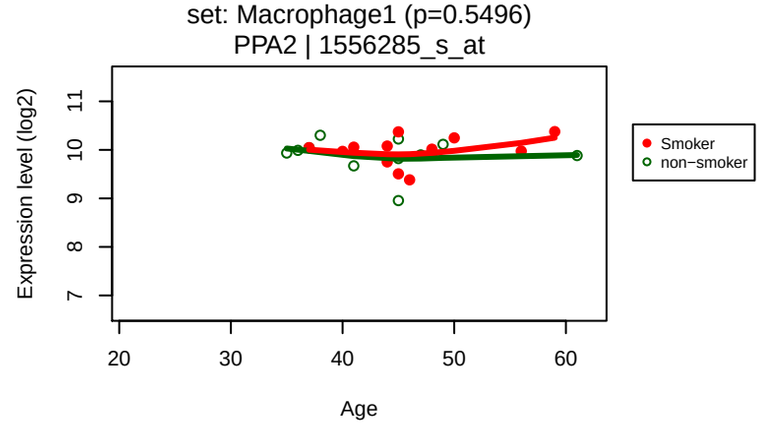

set: SmallAirway1 (p=0.0126)
PPA2 | 1556285_s_at
Expression level (log2)
Age
Smoker
non−smoker

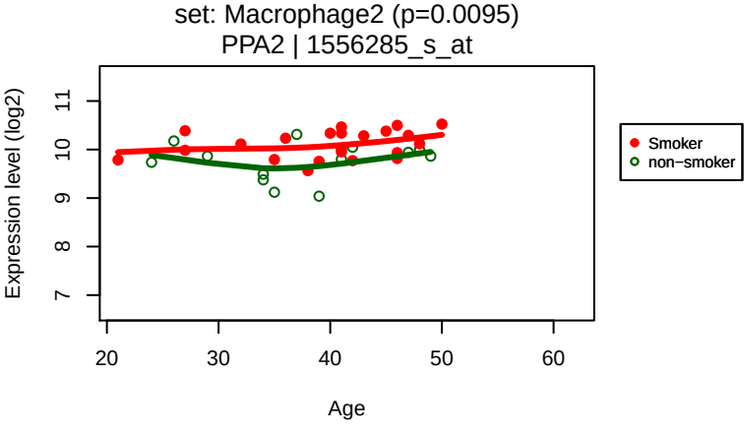

set: SmallAirway2 (p=0.2531)
PPA2 | 1556285_s_at
Expression level (log2)
Age
Smoker
non−smoker

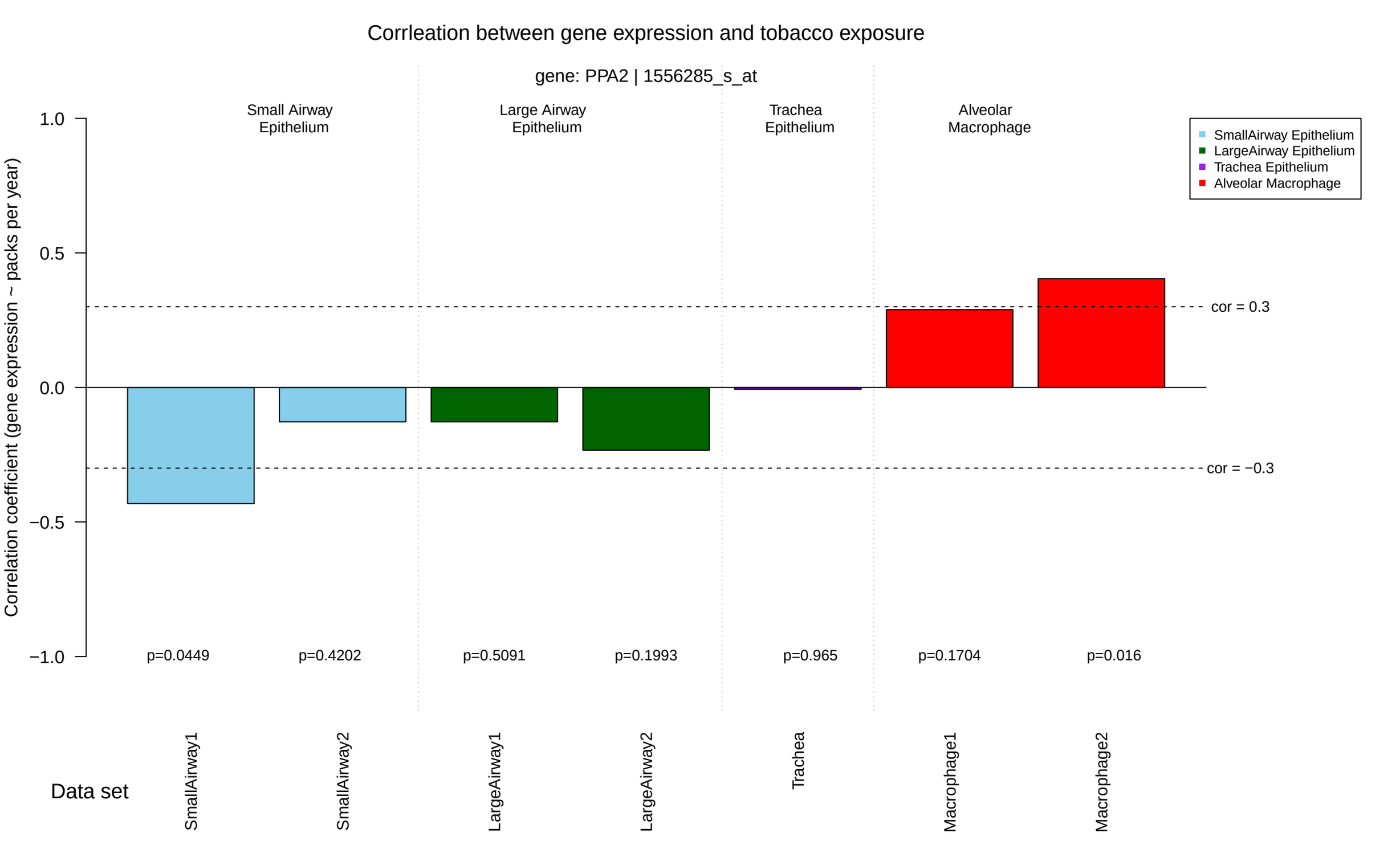

set: LargeAirway1 (p=0.0094)
PPA2 | 1556285_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.366)
PPA2 | 1556285_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8513)
PPA2 | 1556285_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5496)
PPA2 | 1556285_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0095)
PPA2 | 1556285_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPA2 | 1556285_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0449
p=0.4202
p=0.5091
p=0.1993
p=0.965
p=0.1704
p=0.016
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2